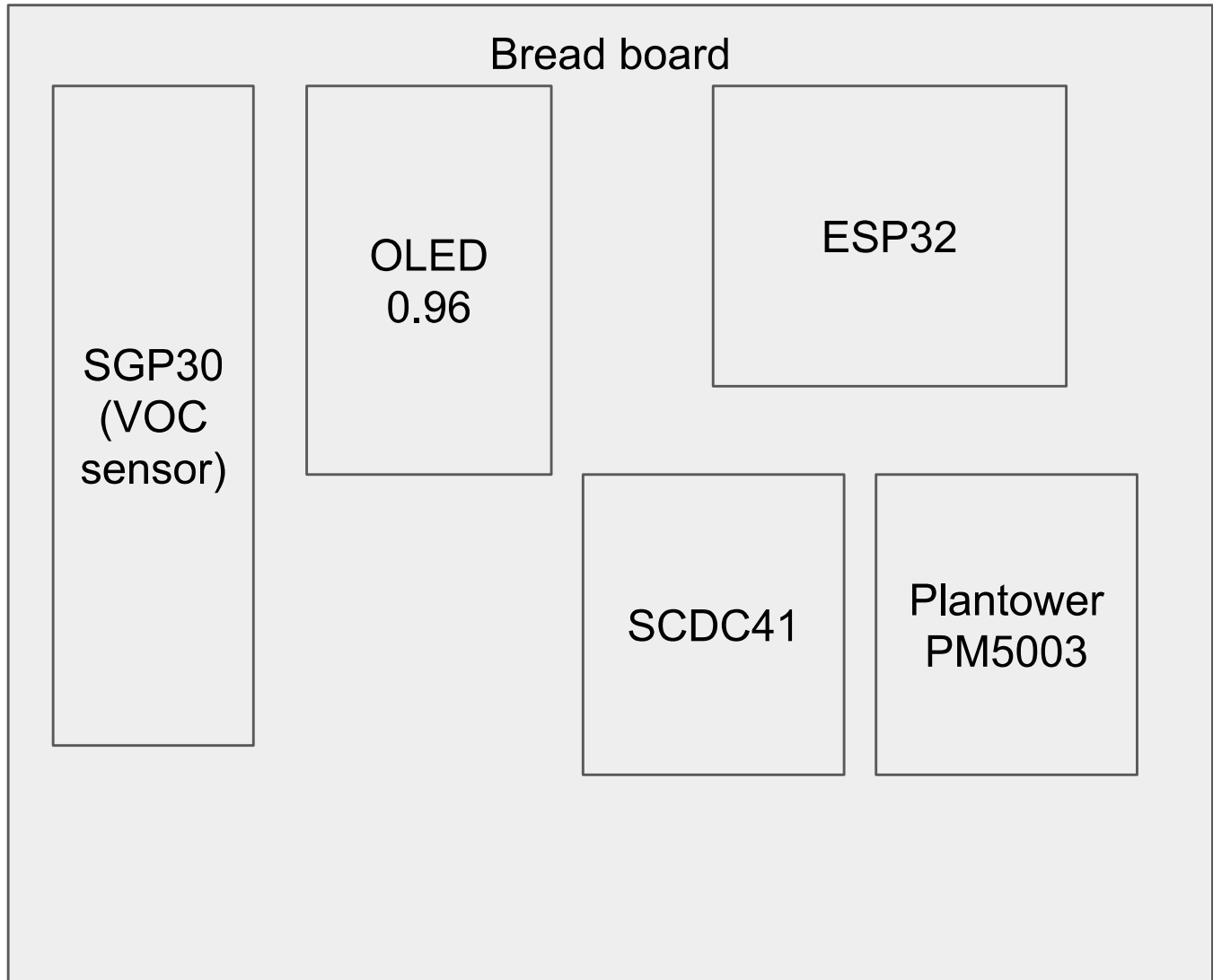


== 3.3V Rail == == GND Rail == == 5V Rail == == GND Rail ==



== 3.3V Rail == == GND Rail == == 5V Rail == == GND Rail ==

Legend:

== = Power Rail (red = positive, blue = ground)

— = Jumper wire connection

★ = Critical voltage difference point

Power rails (Red = + , Blue = -)

**Left side: 3.3V / GND | Right side:
5V / GND**

Flowchart

START

ESP32, SCD41, PM5003, SGP41
Breadboard, power module, jumperwires

Connect 3.3V rails from ESP32
Connect 5V rails from ESP32
Connect Common GND rail

GPIO21 → SDA bus row
GPIO22 → SCL bus row

OLED
SGP41
SCD40

VCC → 5V
GND → GND
TXD → 16

I2D Scanner test

INDIVIDUAL SENSOR TESTS

OLED: "Hello world"

SGP41 VOC reading

SCD41 CO2 reading

PMS5003 PM reading

Combined code

END

AIR QUALITY MONITOR

PM 2.5:  18.7 μg

CO2:  20 ppm

VOC:  30 ppb

TEMP: 23C HUM: 78%
STATUS: GOOD

== 5V Rail == == GND Rail

Bread board

ESP32

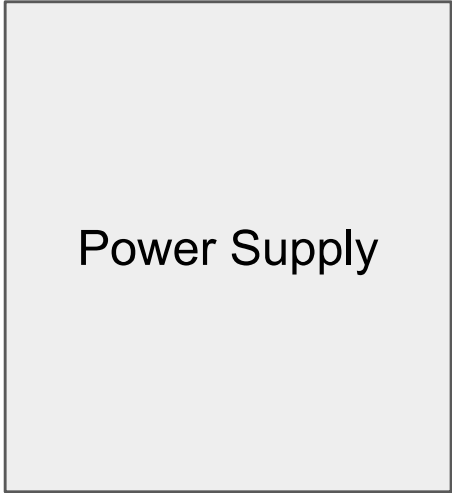
On/OFF

LED lights

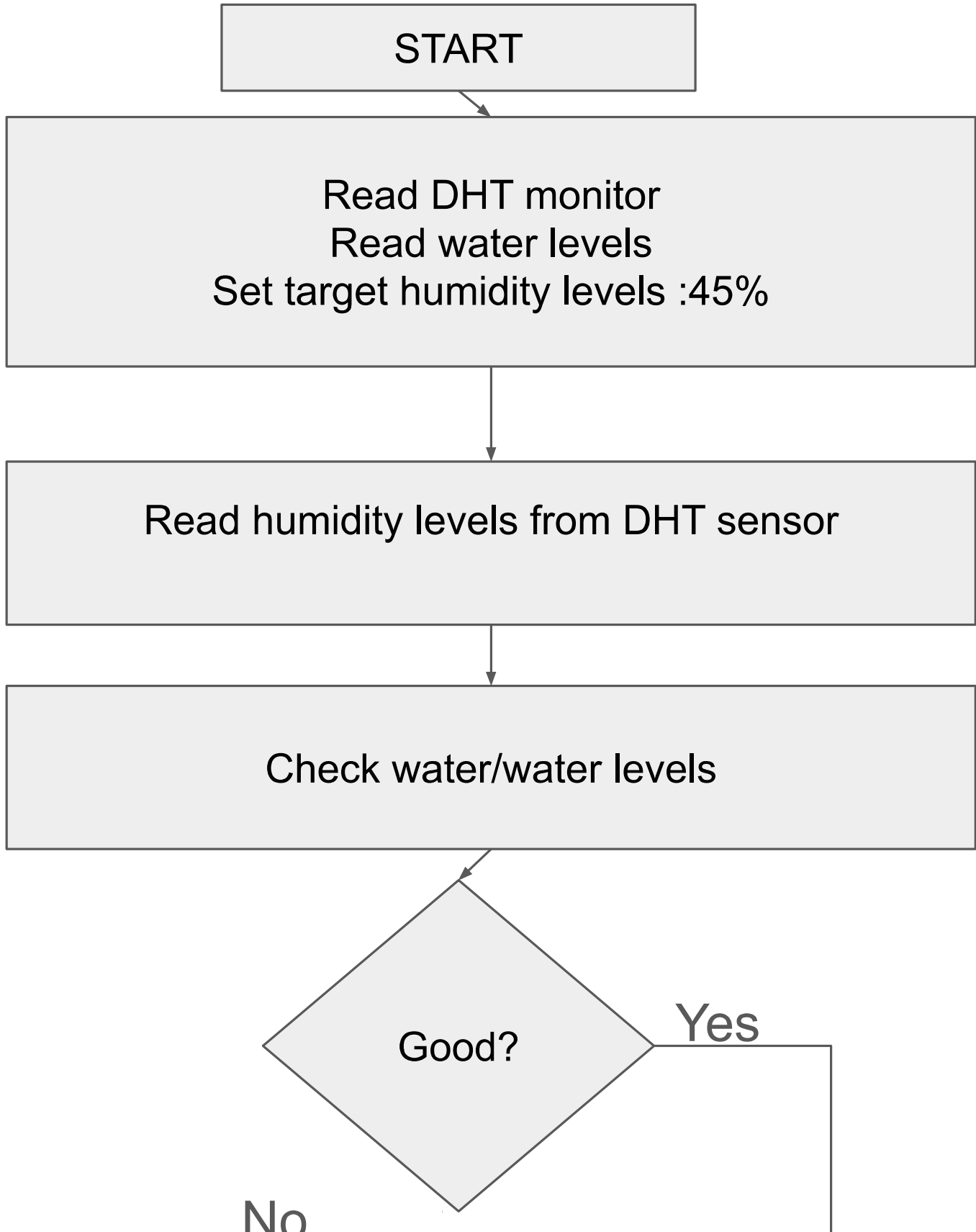
DHT11/22

OLED
(temporary
)

Water level
sensor



Flowchart



No

Yes

Check
humidity

Is it
below
target
?

Yes

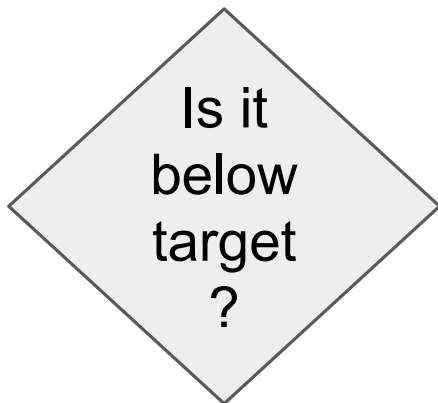
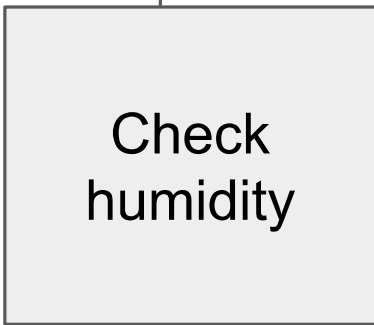
No

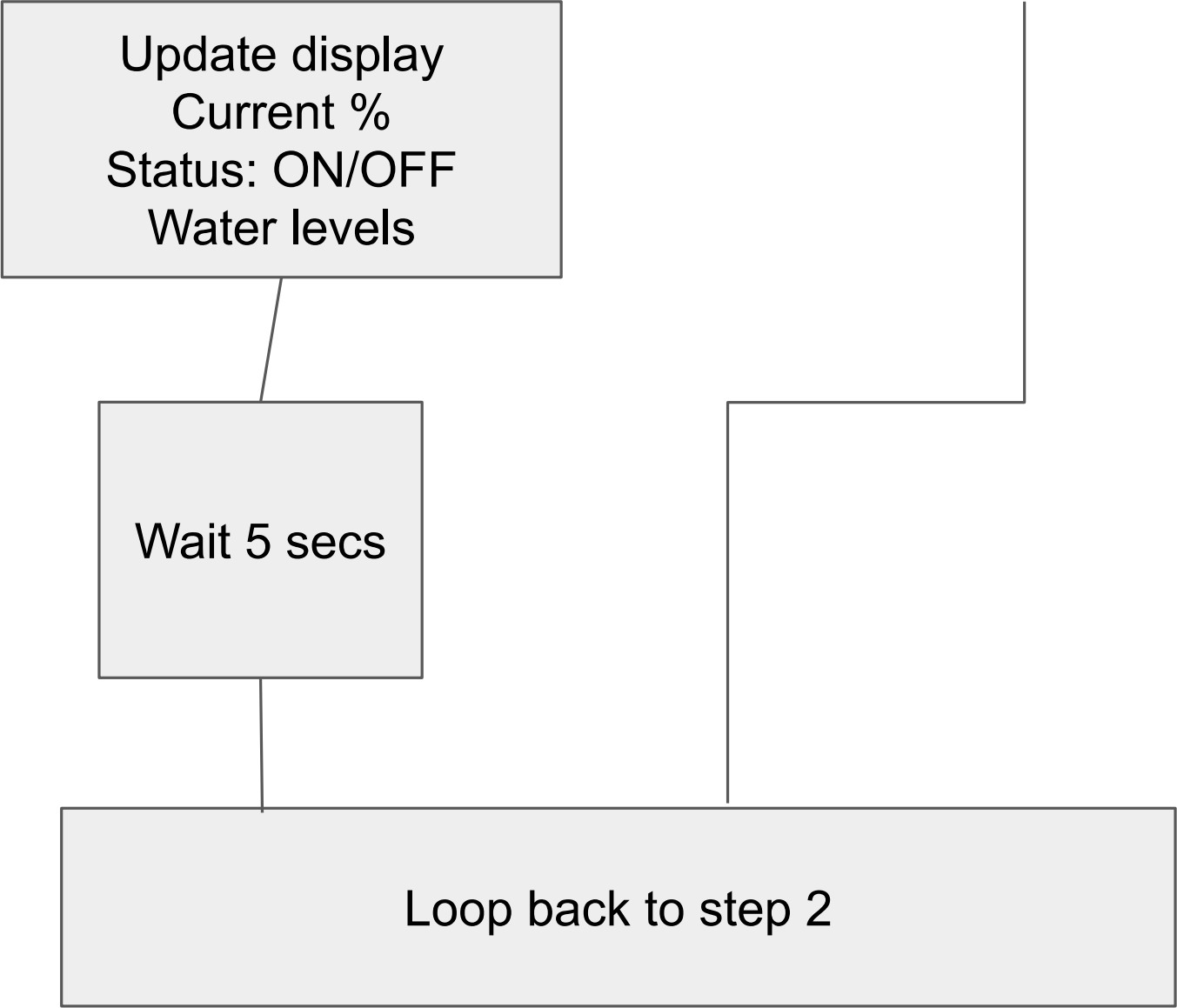
Turn off mist
maker

Turn on mist
maker

Turn off
Green LED

Turn on
Green LED





Notes:

DHT Pin	Connect To	Wire Color
VCC	5V Rail	Red
GND	GND Rail	Black
DATA	Arduino Pin D2	Yellow
<i>(Pull-up resistor)</i>	10kΩ between VCC and DATA	<i>(Required for stable readings)</i>

Step by step wiring

Power rails:

- 3.3V- red rail- red
- 5V- separate red rail- red
- GND blue rail- black

Place DHT11/DHT22 Humidity Sensor

- VCC→ 3.3V rail- red
- GND → GND rail- black
- DATA- GPIO- yellow
-

Place OLED/LCD

– VCC → 5V rail- red

- GND → GND rail - black
- SDA → GPIO 21 - green
- SCL → GPIO 22 - blue

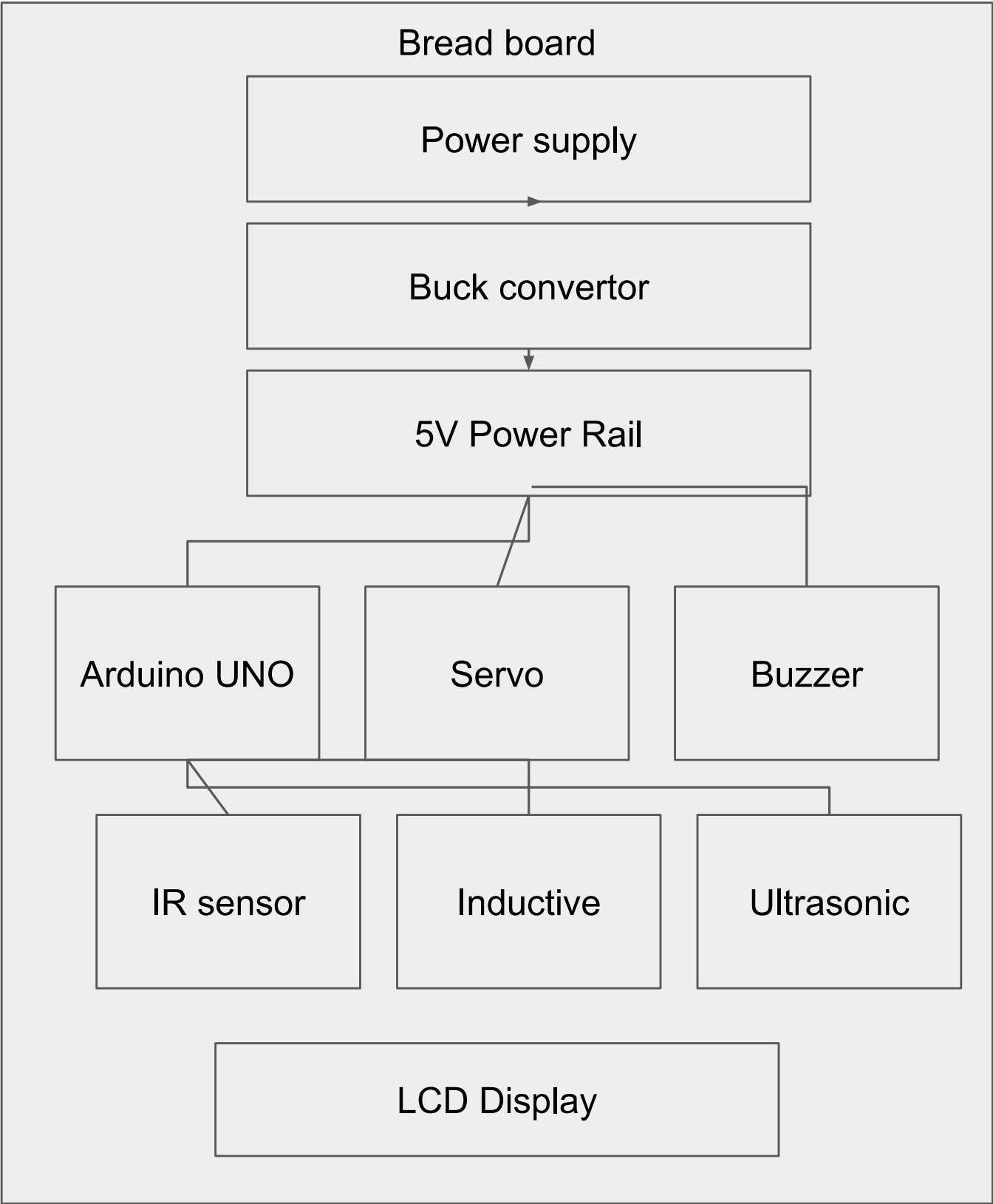
LED

Anode(+) → GPIO 1- white

Cathode (-) → GND rail- black

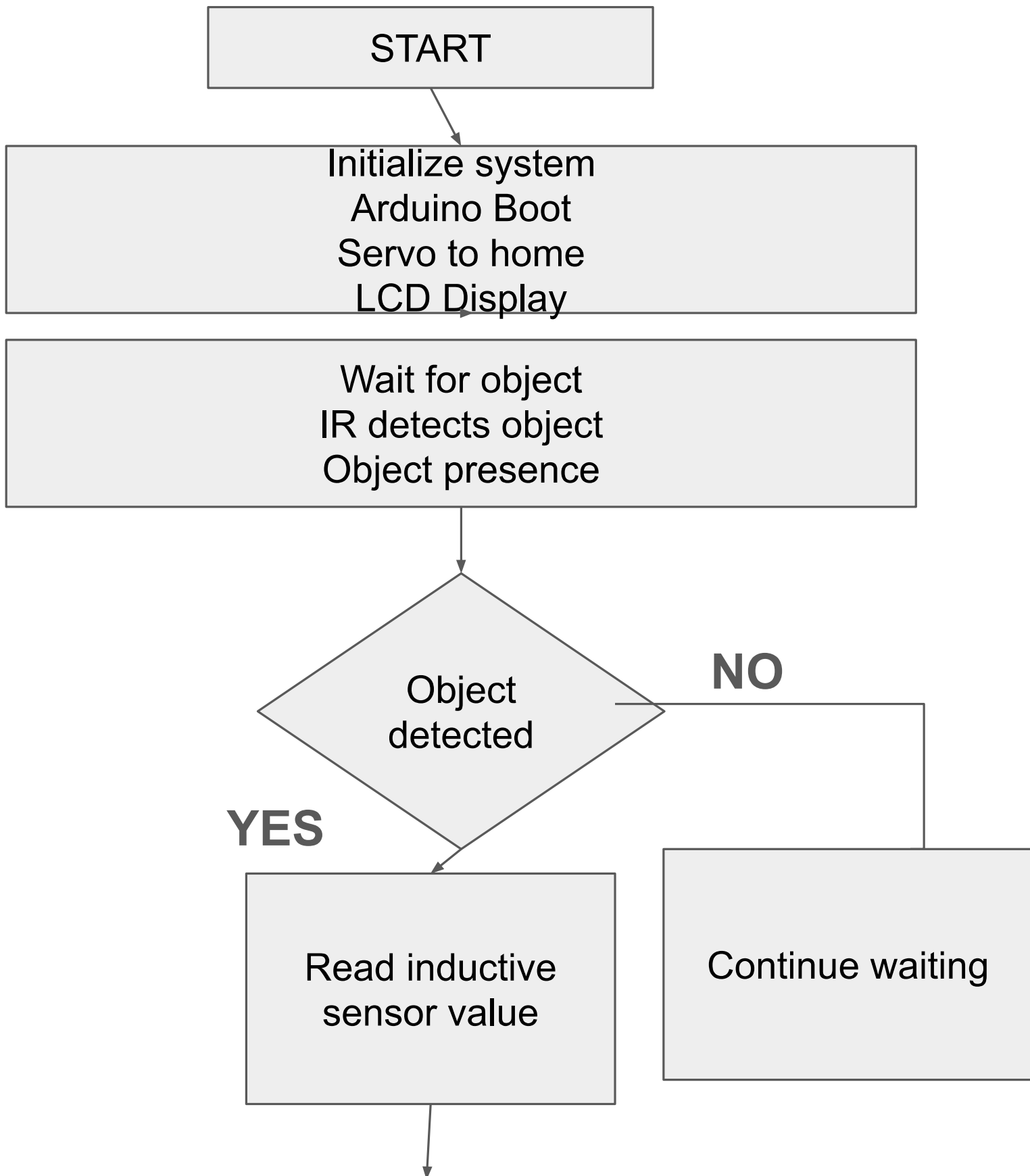
Mist maker and power supply

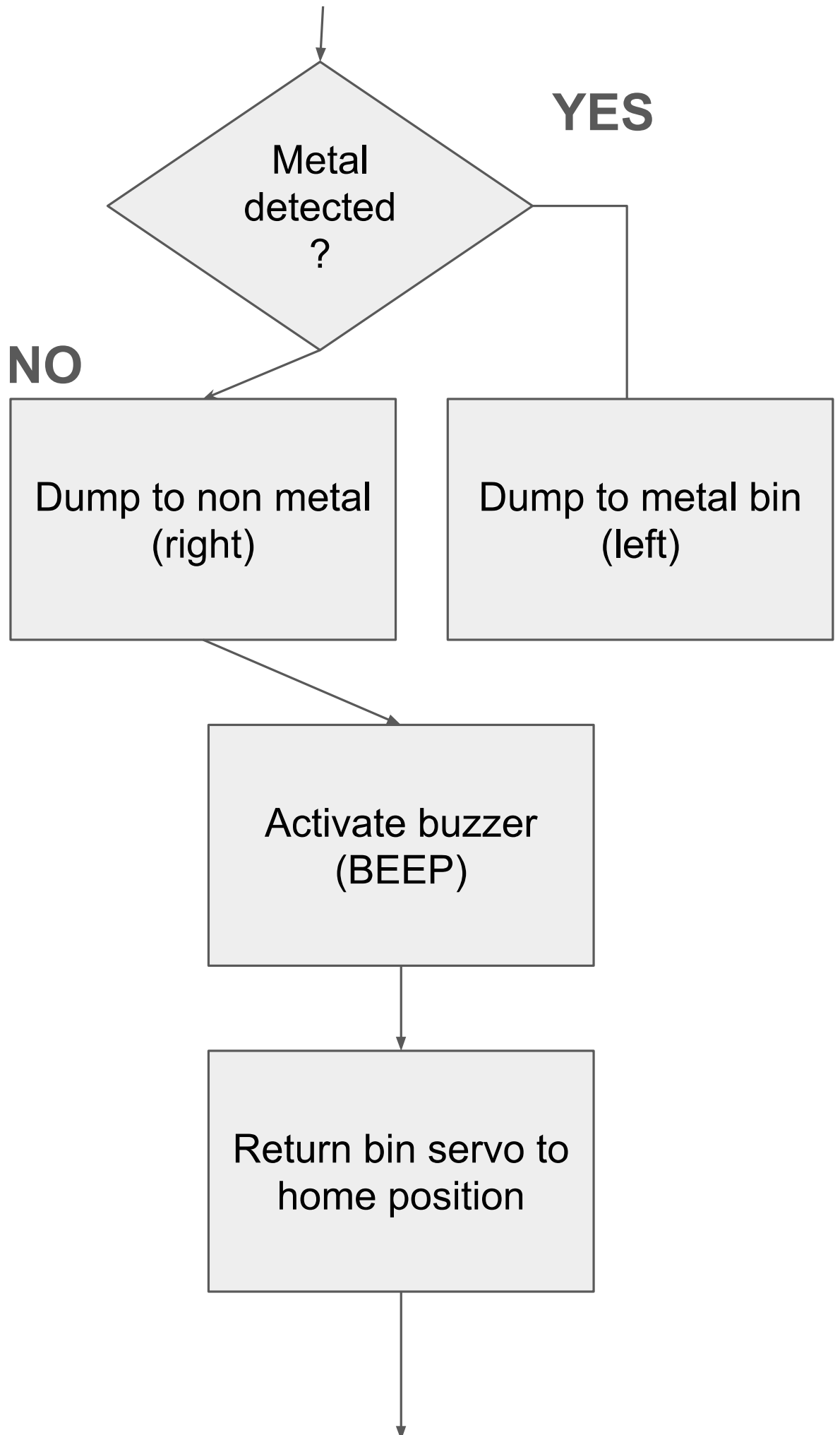
- Connect positive (⊕) to mist maker power supply
- Connect the Normally Open to the Positive mist maker module
- Connect negative of mist maker to negative terminal of mist maker power supply

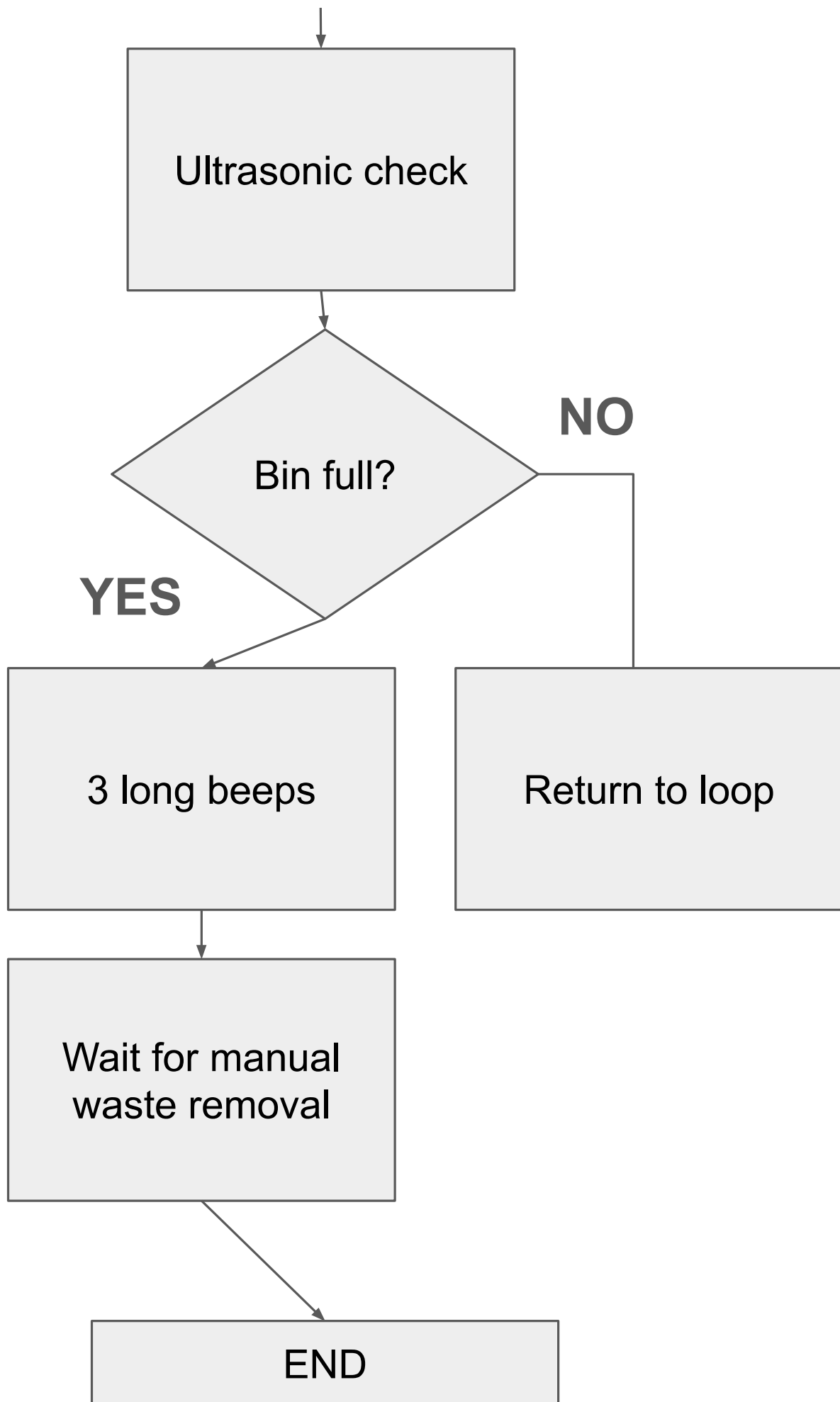


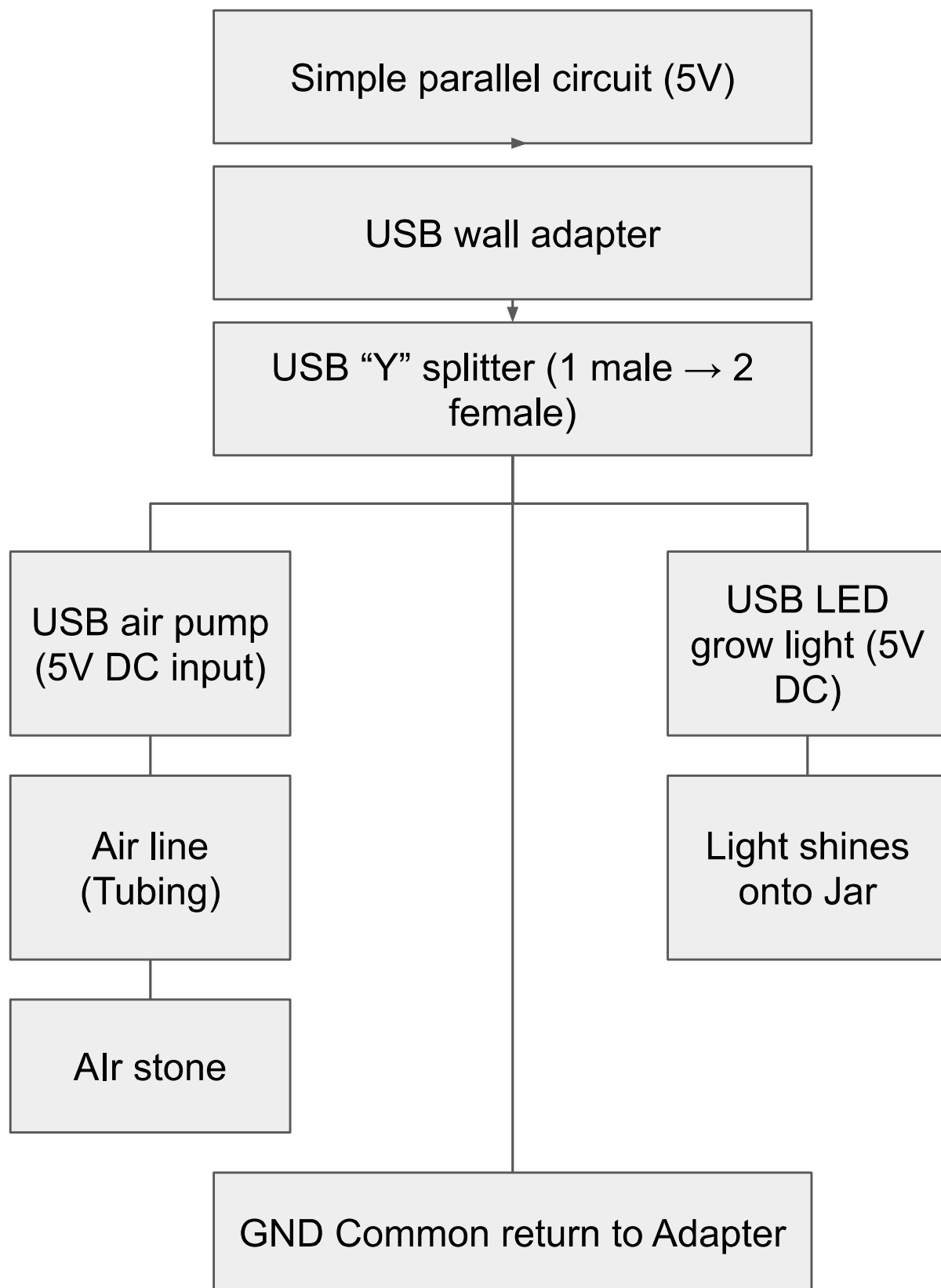
== 3.3V Rail == == GND Rail == == 5V Rail == == GND Rail ==

Flowchart









Flowchart

